

Vaibhav Patil

Backend Developer

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Professional Summary

Backend Engineer with hands-on experience designing and deploying RESTful APIs, user authentication systems, and relational databases. Proficient in Python, Flask, SQLAlchemy, and PostgreSQL in production-grade environments. Applied Machine Learning (ML) concepts across two software engineering internships. Seeking a Python Developer / Backend Developer role to build scalable systems and optimize database performance.

Technical Skills

Languages: Python (Proficient), Java (Intermediate)
Backend: Flask, Node.js, REST APIs, Flask-SocketIO
Databases: PostgreSQL, MongoDB, MySQL
Frontend: HTML5, CSS3, Bootstrap
Tools: Git, GitHub, VS Code
AI/ML: Pandas, NumPy, Machine Learning, Data Preprocessing

Education

Bachelor of Technology (B.Tech) in Computer Science
Pimpri Chinchwad University, Pune 2023 – Present

Work Experience

Python Developer Intern, Pinnacle Labs Jul 2025 – Aug 2025

- Built and shipped backend features for 2 production applications, including an e-commerce platform and real-time chat system using Flask.
- Architected RESTful APIs and integrated PostgreSQL databases using SQLAlchemy ORM, enhancing data access consistency across modules and reducing query times by 20%.
- Implemented authentication, session management, and role-based access control, securing 3 core modules for multi-user workflows.

Artificial Intelligence & Machine Learning Intern, BlackBuck Engineers May 2025 – Jul 2025

- Trained and evaluated machine learning classification and prediction models, achieving 85% accuracy on structured datasets.
- Performed data preprocessing and feature engineering to improve model reliability and reduce noise in input data.
- Delivered ML-driven insights that supported data-based decision-making for the team.

Projects

UrbanCool – AI-Driven Responsive Cooling System

- Designed an AI simulation prototype for urban cooling, integrating weather, IoT sensor, and crowd-density data to propose targeted interventions, reducing predicted energy consumption by 15%.
- Developed a predictive model to flag high-temperature, high-density zones, enabling resource-efficient recommendations over uniform cooling.
- Built an interactive Streamlit dashboard visualizing heat zones and energy use in real time.

ChatApp – Real-Time Chat Application

- Developed a real-time messaging system supporting multiple concurrent users using Flask, Flask-SocketIO, and WebSockets.
- Optimized backend performance to maintain low-latency message delivery under concurrent load.

Certifications

- IBM Machine Learning Certificate (Coursera), 2026
- Red Hat OpenStack
- Walmart Software Engineering Simulation
- Oracle Cloud Data Management

Publications & Achievements

- Presented research paper "AI for Environmental Management: UrbanCool AI Driven Responsive Cooling System" at IEEE I3CTCON 2026.

Additional Information

Languages: English, Hindi, Marathi, German